SE444 – Artificial Intelligence

Makeup Quiz: Logic and Bayesian Networks

First-Order Logic, Knowledge Representation,
Bayesian Networks & Probabilistic Inference

Time: 20 minutes *Total: 10 Questions* *Date: December 22, 2025*

Student Name: _____

Student ID: _____

Section: _____

Instructions

- This quiz contains **10 multiple choice questions**.
- Topics: First-Order Logic (FOL), Knowledge Representation, Bayesian Networks, Probabilistic Inference
- Duration: **20 minutes**
- Some questions may have **multiple correct answers**.
- Write your answers clearly in the boxes provided on the answer sheet below.
- Show your work for computational questions.

Answer Sheet

Fill in your answers below. For questions with multiple correct answers, list all that apply.

Q#	Answer	Q#	Answer	Q#	Answer
1		5		9	
2		6		10	
3		7			
4		8			

Questions

Question 1: FOL Translation

Category: First-Order Logic

Statement: "Every hospital has at least one doctor who can treat all patients admitted to that hospital." Which FOL formula best captures this?

A. $\forall h\,(\text{Hospital}(h) \rightarrow \exists d\,(\text{Doctor}(d) \land \forall p\,(\text{AdmittedTo}(p,h) \rightarrow \text{CanTreat}(d,p))))$

B. $\exists d\,\forall h\,\forall p\,((\text{Hospital}(h) \land \text{AdmittedTo}(p,h)) \rightarrow \text{CanTreat}(d,p))$

C. $\forall h\,\forall p\,(\text{AdmittedTo}(p,h) \rightarrow \exists d\,\text{CanTreat}(d,p))$

D. $\forall d\,\exists h\,(\text{Hospital}(h) \rightarrow \text{WorksAt}(d,h))$

E. $\exists h\,\forall d\,(\text{Doctor}(d) \rightarrow \text{CanTreat}(d,h))$

Question 2: Resolution and CNF

Category: Knowledge Representation

Given the following knowledge base in propositional logic:

1. $(A \land B) \rightarrow C$
2. $C \rightarrow (D \lor E)$
3. $A \land B$
4. $\neg D$

After converting to CNF and applying resolution, what can be inferred?

A. E

B. $\neg E$

C. D

D. $C \land D$

E. The KB is inconsistent

Question 3: Unification and Substitution

Category: First-Order Logic

Given the predicates:

- $P(f(x), g(y), x)$
- $P(f(a), z, a)$

What is the Most General Unifier (MGU)?

A. $\{x/a, z/g(y), y/b\}$

B. $\{x/a, z/g(y)\}$

C. $\{x/a, y/a, z/g(a)\}$

D. $\{x/a, z/g(a)\}$

E. Unification is not possible

Question 4: Bayesian Network Inference

Category: Bayesian Networks

In a Bayesian Network with variables $\{A, B, C, D, E\}$ forming the structure $A \to B \to C$ and $A \to D \to E$, you want to compute $P(E \mid B = b)$. Which variables must be eliminated during Variable Elimination?

A. Only C and D

B. Only A, C, and D

C. Only A and D

D. Only D

E. A, B, C, and D

Question 5: Quantifier Scope and Meaning

Category: First-Order Logic

Which statement is TRUE about the following two FOL formulas?

1. $\forall x \, \exists y \, \text{Loves}(x, y)$ – "Everyone loves someone"
2. $\exists y \, \forall x \, \text{Loves}(x, y)$ – "There is someone loved by everyone"

A. They are logically equivalent

B. Formula (2) implies Formula (1)

C. Formula (1) implies Formula (2)

D. They are both always false

E. Neither implies the other

Question 6: Factor Operations in VE

Category: Bayesian Networks

You have three factors: $f_1(A, B)$, $f_2(B, C)$, and $f_3(C, D)$. You want to eliminate variables in order: B, then C. What is the scope of the final resulting factor after eliminating both?

A. (A, D)

B. (A, B, D)

C. (A, C, D)

D. (B, C)

E. (A)

Question 7: Bayes' Theorem Application

Category: Bayesian Networks

A rare genetic disorder affects 0.1% of the population. A diagnostic test has 98% sensitivity (true positive rate) and 95% specificity (true negative rate). If a randomly selected person tests positive, what is the approximate probability they have the disorder?

A. About 2%

B. About 16%

C. About 50%

D. About 98%

E. Cannot be determined

Question 8: Skolemization

Category: First-Order Logic

Given the formula: $\forall x \, \exists y \, \forall z \, (P(x, y) \rightarrow Q(y, z))$. Which is the correct Skolemized form?

A. $\forall x \, \forall z \, (P(x, f(x)) \rightarrow Q(f(x), z))$

B. $\forall x \, \forall z \, (P(x, y) \rightarrow Q(y, z))$

C. $\forall x \, \forall z \, (P(x, f(x, z)) \rightarrow Q(f(x, z), z))$

D. $\exists y \, \forall x \, \forall z \, (P(x, y) \rightarrow Q(y, z))$

E. $\forall x \, \forall z \, (P(x, f(z)) \rightarrow Q(f(z), z))$

Question 9: d-Separation

Category: Bayesian Networks

Consider a Bayesian Network: $A \rightarrow B \leftarrow C$ (a V-structure with B as the common effect). Which statement is TRUE?

A. A and C are conditionally independent given B

B. A and C are independent (unconditionally)

C. A and C become dependent when B is observed

D. A and C are always dependent regardless of evidence

E. None of the above

Question 10: Complex Inference with VE

Category: Bayesian Networks

In Variable Elimination, which statement about elimination ordering is TRUE?

A. Different elimination orders can produce different final probability values

B. The optimal elimination order always eliminates variables in alphabetical order

C. A poor elimination order can create exponentially large intermediate factors, drastically increasing computation time

D. Evidence variables should be eliminated first

E. Query variables must be eliminated last

— End of Quiz —

Good Luck!